Dahao Tang

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EDUCATION

David James Lab, University of Sydney Mar 2026 – Present
Master of Philosophy, AI & Biology, co-supervised by Prof. David James and Prof. Jinman Kim

University of Sydney Feb 2021 – Nov 2025
Bachelor of Advanced Computing, Computer Science, First Class Honours

University of Sydney Feb 2021 – Nov 2025
Bachelor of Commerce, Finance

PUBLICATIONS

Optimal Look-back Horizon for Time Series Forecasting in Federated Learning AAAI 2026 Oral
Dahao Tang, Nan Yang, Yanli Li, Zhiyu Zhu, Zhibo Jin, Dong Yuan

Stability-Driven CNN Training with Lyapunov-Based Dynamic Learning Rate ADC 2024
Dahao Tang, Nan Yang, Yongkun Deng, Yuning Zhang, Abubakar Sadiq Sani, Dong Yuan

Adaptformer: An Adaptive Multimodal Deep Decomposition Approach for Power Consumption Forecasting ADMA 2024
Nan Yang, Yuning Zhang, Yunqi Wang, Dahao Tang, Yanli Li, Dong Yuan

WORK EXPERIENCE

University of Sydney | *Research Assistant, School of Life and Environment Science* Nov 2024 – Present
Lead developer for ProteHome at the David James Lab; engineering tools to support biological data analysis.

University of Sydney | *Academic Tutor, School of Computer Science* Aug 2024 – Feb 2025
ISYS2120: Data and Information Management (2024 S2), guiding students through database concepts and SQL implementation.

PROJECTS EXPERIENCE

Cytomotif | *Research Project* March 2026 – Present
Develop a web tool for identifying motif patterns in the cytoplasmic sequences of SLC family members that traffic to the plasma membrane in response to insulin.

ProteHome | *Research Assistant, Web Development* July 2024 – Present
Spearheaded website refactoring and architected the full-stack solution using Next.js, Django, and PostgreSQL. Implemented Tailwind CSS to modernize the UI and continuously contributed to feature development and optimization.

ioulia.ai | *Co-founder* Jan 2024 – July 2024
Designed the technical roadmap and engineered the Minimum Viable Product (MVP) from scratch. Optimized product performance and LLM responses by implementing advanced prompt engineering techniques such as Chain-of-Thought (CoT).

GigHero | *Co-founder* Apr 2023 – Aug 2024
Developed a two-sided online marketplace MVP for on-site services using React and Express.

Searten | *University Project, Project Manager, Full-stack Developer* Aug 2023 – Nov 2023
Led a cross-functional team to develop a web-based platform MVP, overseeing the project lifecycle.

File System Implementation with FUSE | *University Project* 2022
Implemented a functional file system in C using the FUSE interface, engineered support for large file handling and hierarchical directory structures.

OS Scheduler Implementation | *University Project* 2022
Designed and implemented an operating system scheduler in C. Integrated Round Robin and Multi-Level Feedback Queue (MLFQ) algorithms to optimize process execution efficiency.